Mascot for AutoReCOrd (M.A.R.C.O.)

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Table of Contents

Mascot for AutoReCOrd 1

I. Project Synopsis. 3

II. Architecture. 3

a. The Mechanics of the Software 3

1. Summary 3

Distinguishing Features 3

2. Constraints 4

3. Use Cases 4

III. User Interface Design Draft. 5

IV. Database Schema 6

V. High Level Diagram 6

I. Project Synopsis.

Mascot for AutoReCOrd is a web application that schedules tasks with no fixed timeslot, such as grocery shopping, laundry, studying, etc., for users.

II. Architecture.

a. The Mechanics of the Software

1. Summary

Mascot for AutoReCOrd (M.A.R.C.O.) is a web application that is optimized for use on mobile devices with the main goal of aiding the user with *tracking, scheduling,* and *editing* flexible tasks such as laundry, grocery-shopping, and study sessions. Additionally, users can create custom tasks.

M.A.R.C.O. will offer an *initial estimation* of how long each task should take based on the mean time all other users took to complete the same task.

As the user records how long each task should take (recorded either via a timer or manual entry), the estimation will adjust to reflect the individual user's mean time to complete the task.

Distinguishing Features

M.A.R.C.O. offers the capability of interacting with *Google Calendar*, allowing the users to schedule their events around existing schedules within the application.

M.A.R.C.O. will come with a small virtual assistant that will remind the user to start or stop the task tracking timers based on past data. Whenever a timer is left running for irregularly long periods of time, or when a user might've forgotten to start

the tracker for an already scheduled task, the virtual assistant will remind them to start or stop the timer.

M.A.R.C.O. will offer a tailored experience for mobile devices using an installable *Progressive Web App* (PWA for short), with some UI adaptations and functionality being adjusted for better mobile device use.

2. Constraints

i. All app functions should be accessible and provide a good user experience on mobile devices, specifically smartphones.

ii. Integration with *Google Cloud Services* and *Google Calendar API*.

iii. The assistant should not be intrusive and should only serve as mitigation to reduce inaccurate records introduced by user error.

3. Use Cases

This software is a web application to help users schedule and track flexible tasks, which are tasks that can be completed at any non-fixed time.

Examples of flexible tasks include laundry, grocery shopping, and course work.

Users can import events from their Google Calendar and add new fixed time tasks within the web application itself.

The user can input a hard deadline and priority level for each flexible task, as well as an initial estimation of time necessary to complete the task.

III. User Interface Design Draft.

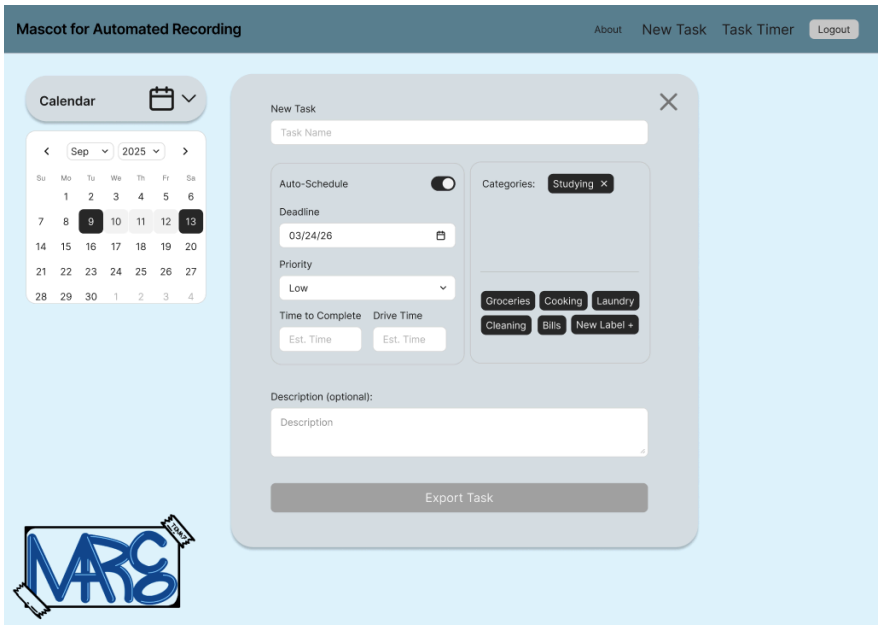

Figure 1. *User interface draft for exporting a task. Made in Figma.*

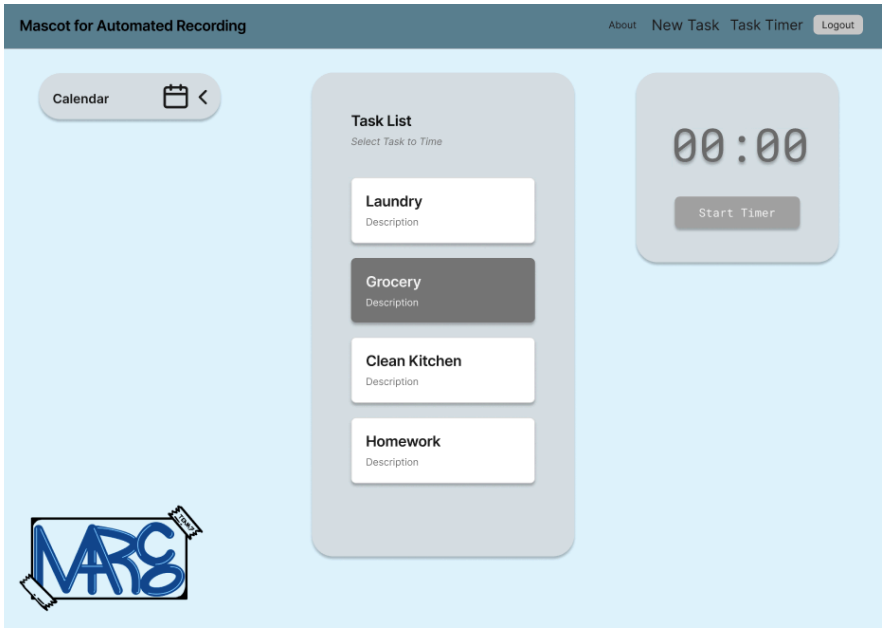

Figure 2. *User interface draft for task timing. Made in Figma.*

IV. Database Schema

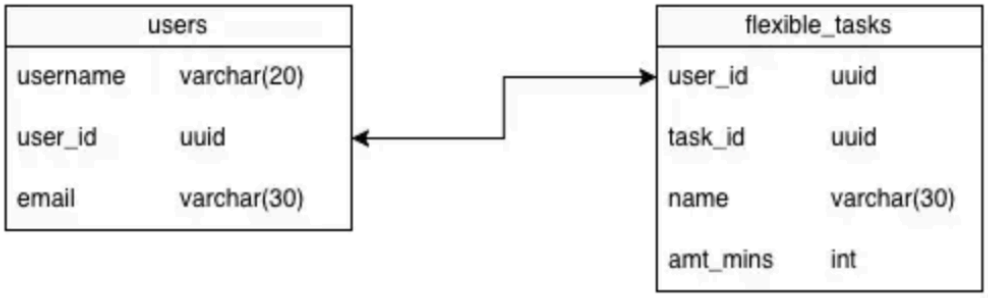

Figure 3. *Diagram of the system's database schema.*

V. High Level Diagram

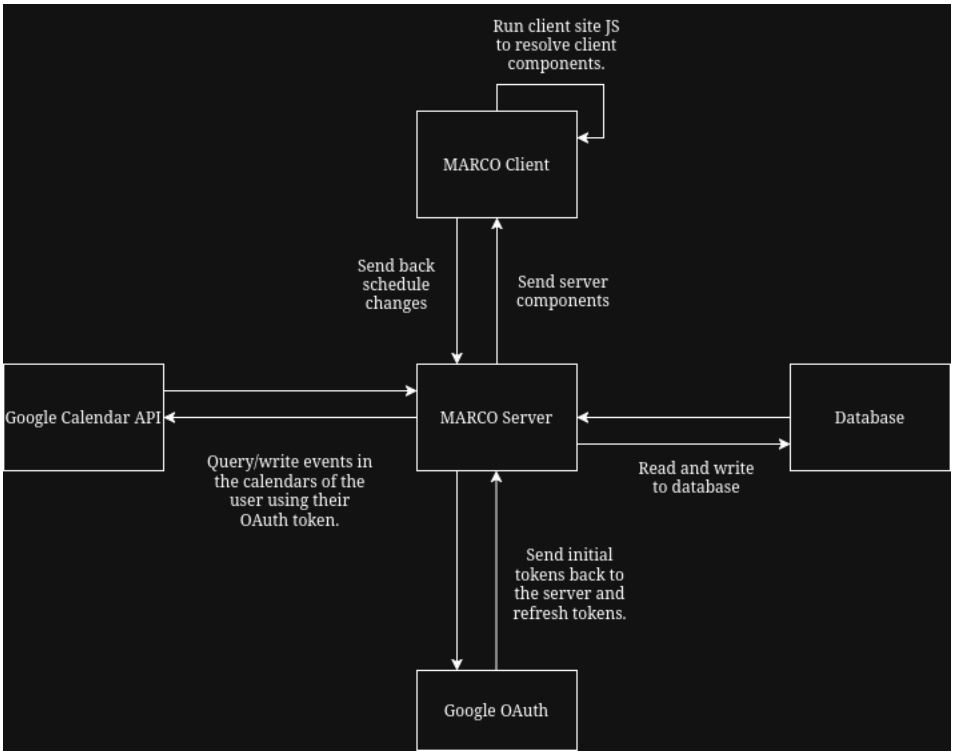

Figure 4. *High level diagram of the system.*